

Sensing Module Prototype for Subsurface-level Wildfire Monitoring

1 Summary

This document presents a comprehensive overview of the design process for a sensing module developed for subsurface wildfire monitoring. The process began with the derivation of specifications from identified needs, goals, and objectives (NGOs), and progressed through system design, fabrication, integration, and verification. The final prototype successfully demonstrated the intended core IoT functionality. A timeline of the design process is provided below, highlighting key artifacts at each stage. The remainder of this document is structured in accordance with this timeline; readers are encouraged to use the embedded artifact links to navigate between supporting materials. Please visit Design Concept before going through this document.

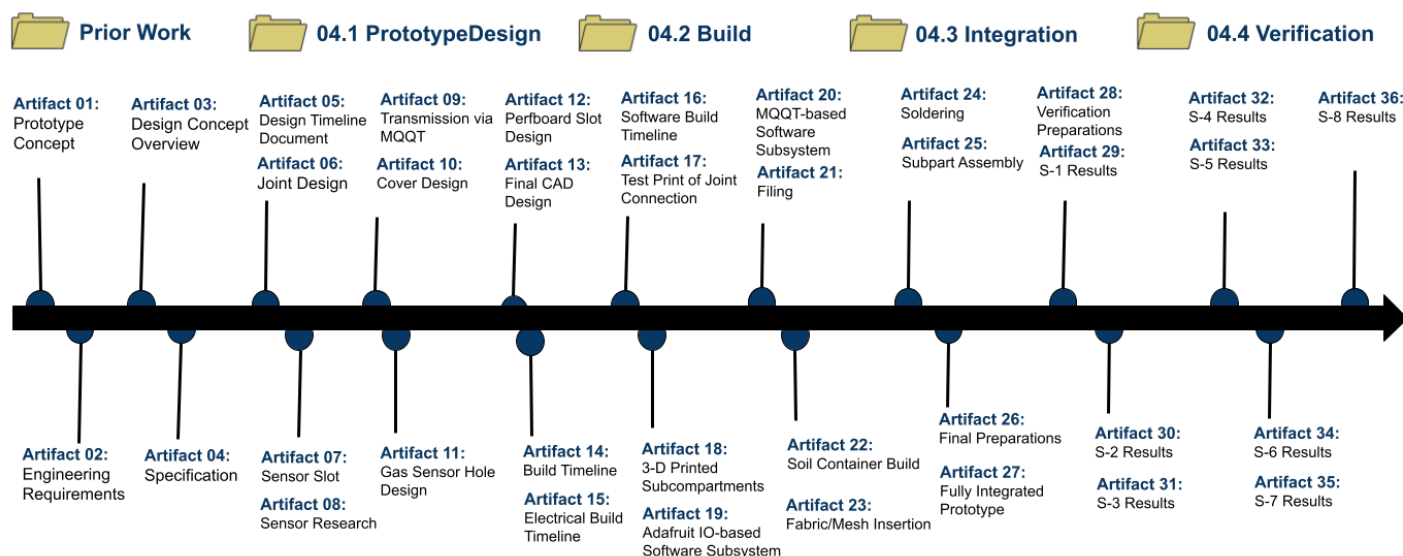


Figure 1: Comprehensive timeline of design, build, and integration, alongside key links to all artifacts. Artifacts 05,14,15,16, provide even more detailed timelines to the precise day. All links below are clickable:

Links: 01, 02, 03, 04, 05, 06, 07, 08, 09, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36.

2 Specification

A set of specifications (S) was first derived from the identified needs, goals, and objectives (NGOs), with a primary focus on the device's IoT functionality (discussed in detail in Section 3). These NGOs were translated into clear, measurable specifications that guided the design. The complete set of NGOs can be found in Requirements.

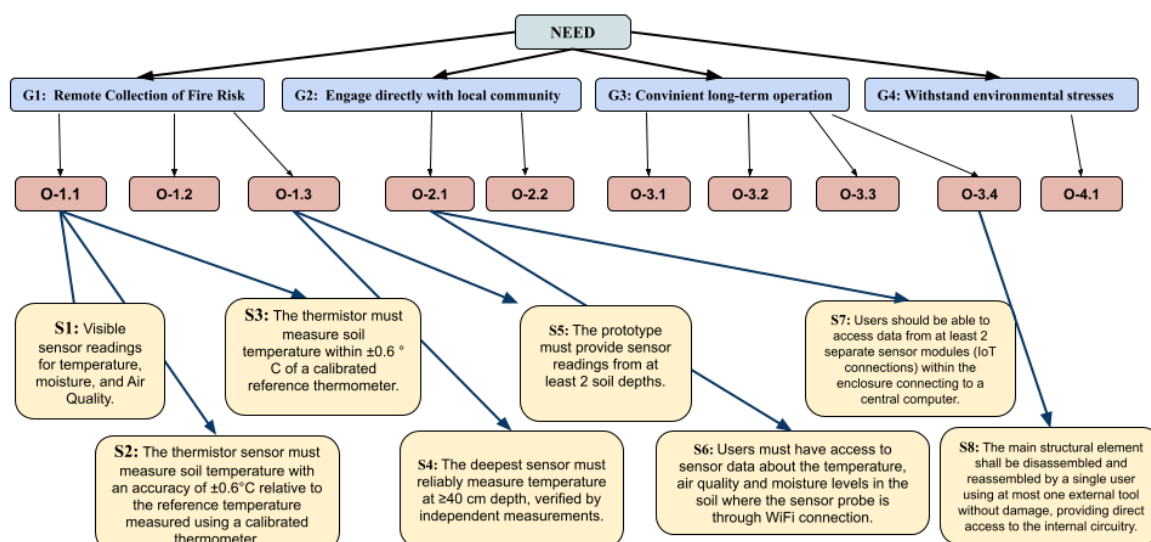


Figure 2: The specification document outlines the intended functionality of the prototype system.

3 Summary of Design Process

In Design Dossier 1, we converged on an auger-based design that incorporated a spiral screw mechanism to facilitate efficient insertion into peatland soils. Since we seen these devices used before for digging soil, we elected to prototype the more complex sensing subsystem, implementing the sensors on a simplified elongated probe rather than integrating the full auger structure. See the Design Overview for more information on the design. We started by ensuring our design could be constructed in MyFab.

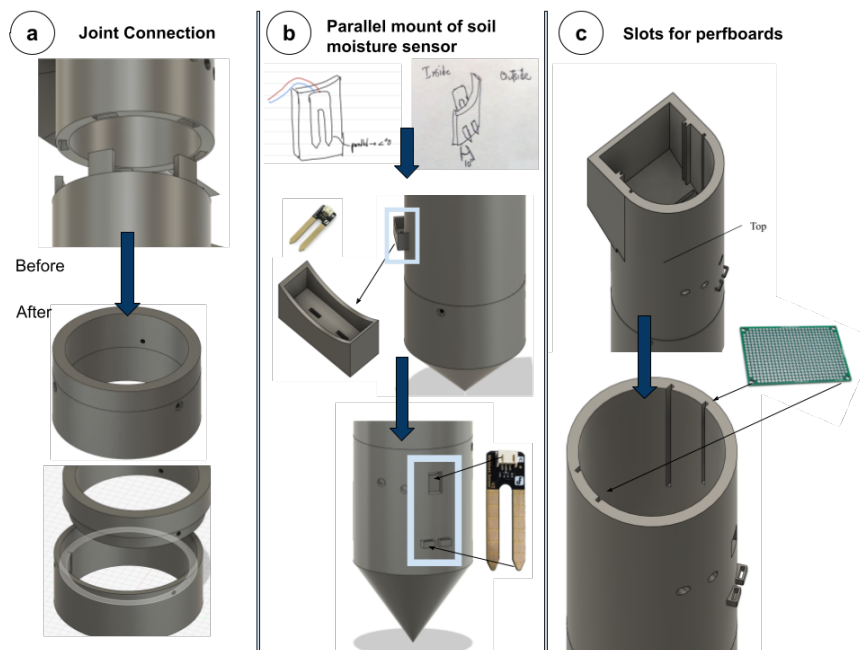
3.1 Design (Click: Design Timeline)

3.1.1 Structural Design

Below are several modifications to our initial design due to MyFab and practicality constraints.

(a) **ARTIFACT 06** Figure 3 a): The first modification arose from Specifications 4 and 5, which require temperature readings from depths greater than 40+ cm underground. However, MyFab's Prusa MK4 has a maximum print height of 22 cm. As a result, the structure had to be divided into three compartments that could be securely fastened together. We converged on a sleeve with a cross-pin mechanism to ensure a tight seal while also allowing for easy disassembly (S8). Click here for full process - pg.4

Figure 3: RIGHT: a) Joint Connection Process, b) Parallel mount of SKU SEN0144, c) Slots for Perfboards



b) **ARTIFACT 07** Figure 3 b): The placement of the SKU SEN0114 soil moisture sensor was another key design challenge (as required by S1,S2,S4 and S5). The sensor needed to be exposed to the peat while not interfering with the ability to screw the device into the ground. Additionally, the internal circuitry could not be exposed to moisture in the soil. To minimize resistance and maintain functionality, the sensor was ultimately mounted as shown in Fig. 3b. This configuration provides space for internal circuitry while positioning the sensor to reduce electrical resistance. **Description of Sensor Choice (p. 1-7)**

c) **ARTIFACT 12** Figure 3 c): The use of internal slots for perfboard mounting was selected to balance secure attachment with ease of access. Alternative approaches, such as adhesives or fixed mounts, were rejected due to limited maintainability and difficulty in accessing internal components. The slot-based design allows perfboards to be easily inserted and removed while remaining securely positioned, supporting modularity and efficient integration of the electrical subsystem.

Other structural design decisions included the addition of a cover for the module (**ARTIFACT 10**), as well as the incorporation of gas sensor holes: **ARTIFACT 11**. Annotated images are provided in the corresponding folder. The final CAD Design is shown below: Click here for more detail.

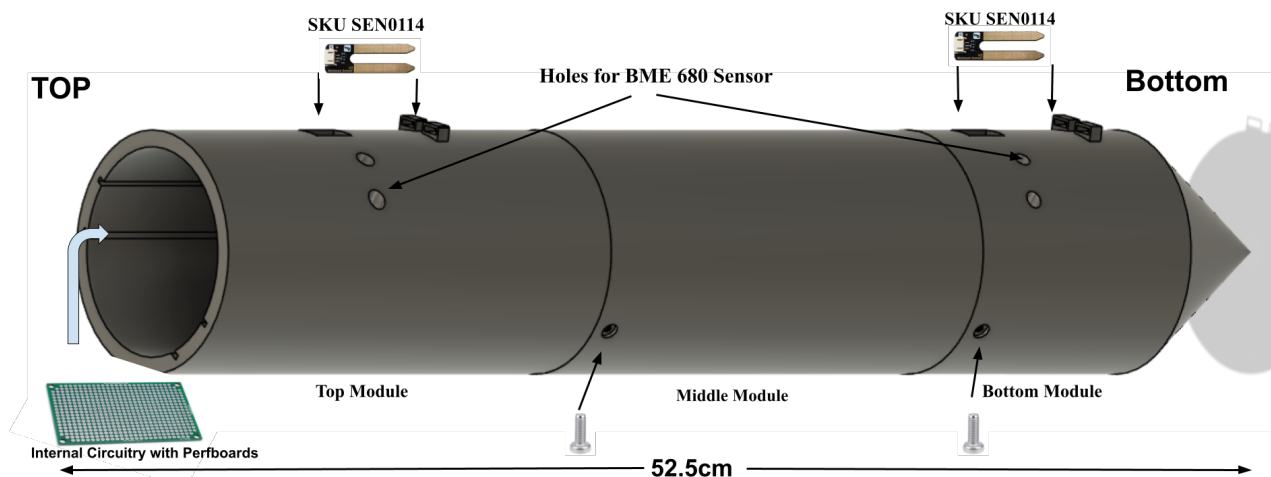


Figure 4: The specification document outlines the intended functionality of the prototype system. This implements all of the changes from Artifacts 5-13.

3.1.2 Software and Hardware Design (Software Timeline)(Electrical Timeline)

Alongside the structural design decisions, the team developed the hardware and software subsystems. The system architecture and circuit diagram are shown below.

ARTIFACT 08:

The final prototype utilizes the BME680 and SKU: SEN0114, selected from an initial suite that included the AM2320 and Grove MQ2. The AM2320 was removed as its capabilities were redundant to the BME680 and it required additional resistors and cabling. The Grove MQ2 was also excluded because its analog output offered less precision for air quality monitoring than the BME680 gas resistance measurements.

ARTIFACT 09: A major design decision was the switching from the Adafruit IO "Feed" interface to a Message Queuing Telemetry Transport (MQTT) protocol-based solution. This was due to the restrictive Adafruit IO bandwidth and data structure requirements, and due to MQTT being the standard for IoT networks. MQTT was chosen, as it is a lightweight architecture, relying on publishers (senders of data), brokers, and subscribers (receivers of data). Separation of concerns can also be done through the topics published to.

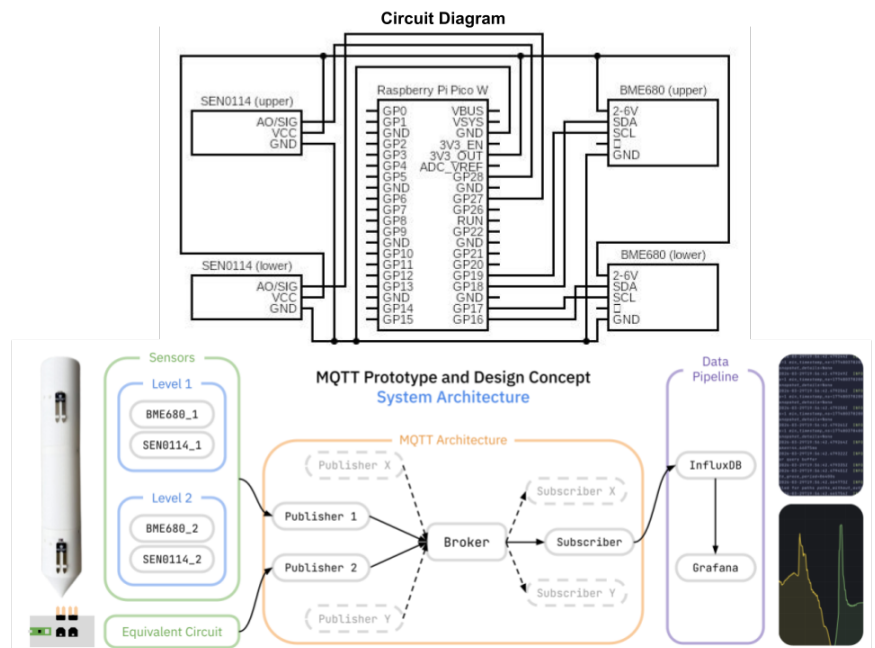


Figure 5: (Top) Circuit diagram for the final iteration of the device. (Below) Software subsystem architecture. Arrows indicate direction of data transfer. Dashed lines indicate components not part of the prototype but part of the design concept, present in the MQTT architecture, the most scalable component of the prototype. Data is gathered by the sensor modules, relayed to the publisher, relayed to the broker, relayed to subscribers, then piped through InfluxDB, and visualised in Grafana

3.2 Build

We conducted a methodical building procedure, beginning with small 3D printed tests of joints.

3.2.1 Structural Build (Click: Timeline)

ARTIFACT 17: Figure 6 (a–b): The sleeve with cross-pin mechanism (Section 3.1.1) was prototyped and tested with the screw interface. The assembly functioned as intended, validating the selected design approach.

ARTIFACT 18: The initial 3D print resulted in filled joints (c). These were removed and refined (d), producing the final components shown in (e). All joints were tested to ensure the perfboard, screws, and sensors could fit.

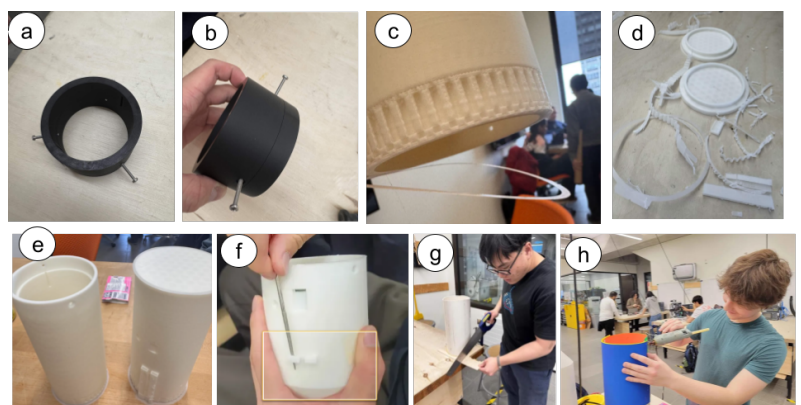


Figure 6: Structural build of the prototype

ARTIFACT 21: An extensive filing process was carried out to ensure that the joint holders for the SKU SEN0144 and the openings for the BME680 sensor were fully open (f). It was now fully prepared for integration.

ARTIFACT 22: A soil container was constructed using MyFab scrap materials (g), with a diameter of 15 cm to fully encompass the sensor module. A height of 55 cm ensured complete vertical enclosure, with additional material added at the top to achieve the required soil depth (h).

3.2.2 Software and Hardware Build (Click: Software Timeline) (Click: Electrical Timeline)

ARTIFACT 19: The code from the in-class IoT workshop was repurposed to send Adafruit IO Feeds for each sensor (BME680,

SEN0114) reading. After the implementation of all sensors, the circuitry was finalised. After functionality was satisfactory, the code was broken up into modules with a focus on separation of concerns (Wi-Fi connection, Adafruit IO interfacing, sensors, configuration).

ARTIFACT 20: The Adafruit IO API was found to be insufficient for the volume of data that had to be transmitted and visualised. Following the advice of Prof. Asare (as per Artifact 09), the software subsystem was drastically refactored to use the MQTT protocol, using an online broker provided by HiveMQ. Data injection and visualisation leveraged InfluxDB and Grafana, respectively. Functioning publisher and subscriber side code were pushed to GitHub (subscriber repository, publisher repository).

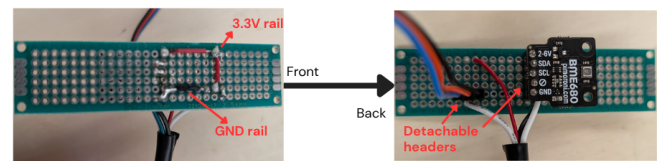
3.3 Integration (Click: [Integration Timeline](#))

ARTIFACT 23 Firstly, to prevent soil and excessive water ingress from entering the BME680 gas holes, a MyFab cloth was cut and secured over the gas sensor openings, allowing gas flow while restricting water penetration.

ARTIFACT 24:

To interface the electrical and structural subsystems, soldering was used. Different techniques were used to promote modularity and simplicity. For the sensor perfboards (see top), both the BME680 and SEN0114 required 3.3V and GND, so these were each combined in parallel to reduce the number of wires going between the sensor and central perfboards. Headers were also soldered to the perfboard (see right), to ensure all components were removable. This made exchanging damaged modules easy.

1. Secondary (sensor) perfboards



2. Main (central) perfboard

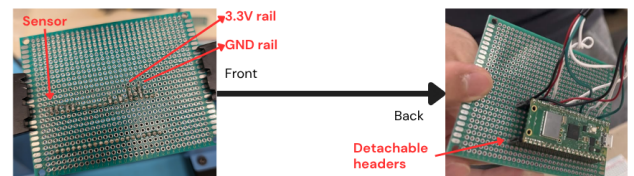
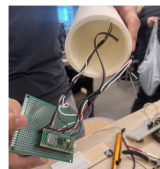


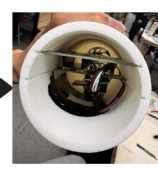
Figure 7: Soldering Procedure of main and secondary perfboards

ARTIFACT 25: Figure 8 shows the step-by-step assembly of the sensing module. First, cables are routed to organize electrical connections and reduce interference. Next, the circuit is mounted onto a perfboard and placed within the housing, followed by securing the soil sensors in position for proper measurement. Finally, the sections are screwed together to complete the structural assembly.

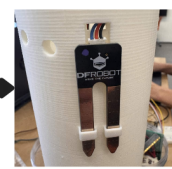
1. Cable routing



2. Perfboard mounting



3. Soil sensor mounting



4. Screwing sections



Figure 8: Sub-component assembly flowchart illustrating the build process.

ARTIFACT 26: Prior to verification on March 29, several issues were identified and resolved. The screw connection was reinforced with hot glue, and an opening was drilled in the top cover to allow USB access for powering the device. These modifications were completed promptly, enabling verification to proceed as planned.

4 Performance of Design

After constructing the testing apparatus and device, eight tests were conducted to evaluate all specifications in Section 2. These tests were designed to address each specification and objective, verifying: (1) integration of multiple sensor layers within a single prototype, (2) IoT connectivity across sensors, and (3) the system's ability to detect variations in temperature, moisture, and air quality.

ARTIFACT 28:

To prepare for verification protocols involving placing the device in soil, electrical tape was applied to the top of the sensors to prevent minor water ingress (a). The test setup consisted of: (i) an Amprobe TMD-30 thermometer inserted into the soil, (ii) the soil testing apparatus, (iii) the fully assembled probe, and (iv) an independent circuit from a secondary device used to verify IoT connectivity. The thermometer was used for calibration of the sensors, specifically for S2.

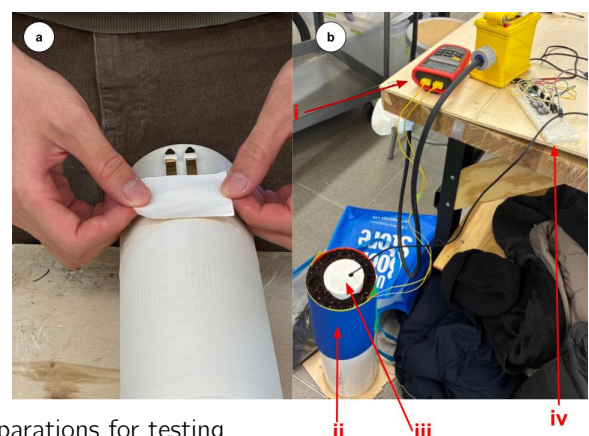


Figure 9: Diagram of final preparations for testing

See [Verification Protocol](#) for the precise experiments. These experiments were all conducted on March 28, 2025 in MyFab.

Table 1: Verification Protocol Results Summary

ID	Specification Statement	Video Proof	Status
S-1	The prototype must provide visible sensor readings for soil temperature, moisture, and Air Quality levels.	Artifact 29	✓ Passed
S-2	The thermistor sensor must measure soil temperature with an accuracy of $\pm 1^{\circ}\text{C}$ relative to a calibrated thermometer.	Artifact 30	✓ Passed
S-3	The moisture sensor must detect an increase in soil moisture when the soil is watered.	Artifact 31	✓ Passed
S-4	The deepest sensor must provide readings at a depth of ≥ 40 cm below the surface.	Artifact 32	✓ Passed
S-5	The prototype must provide sensor readings from at least 2 soil depths.	Artifact 33	✓ Passed
S-6	Users must have access to sensor data (temp, air quality, moisture) through WiFi connection.	Artifact 34	✓ Passed
S-7	Users should be able to access data from at least 2 separate sensor modules (IoT) to a central computer.	Artifact 35	✓ Passed
S-8	The main structural element shall be disassembled and reassembled by a single user using at most one tool.	Artifact 36	✓ Passed

Some key challenges/takeaways from verification were that: (1) it was significantly more difficult to alter the soil temperature at deeper layers, particularly after saturation with cold water. As a result, most temperature changes were observed near the surface, establishing a clear gradient from top to bottom; and (2) the soil required approximately 5–15 minutes to reach thermal equilibrium.

5 Final Prototype

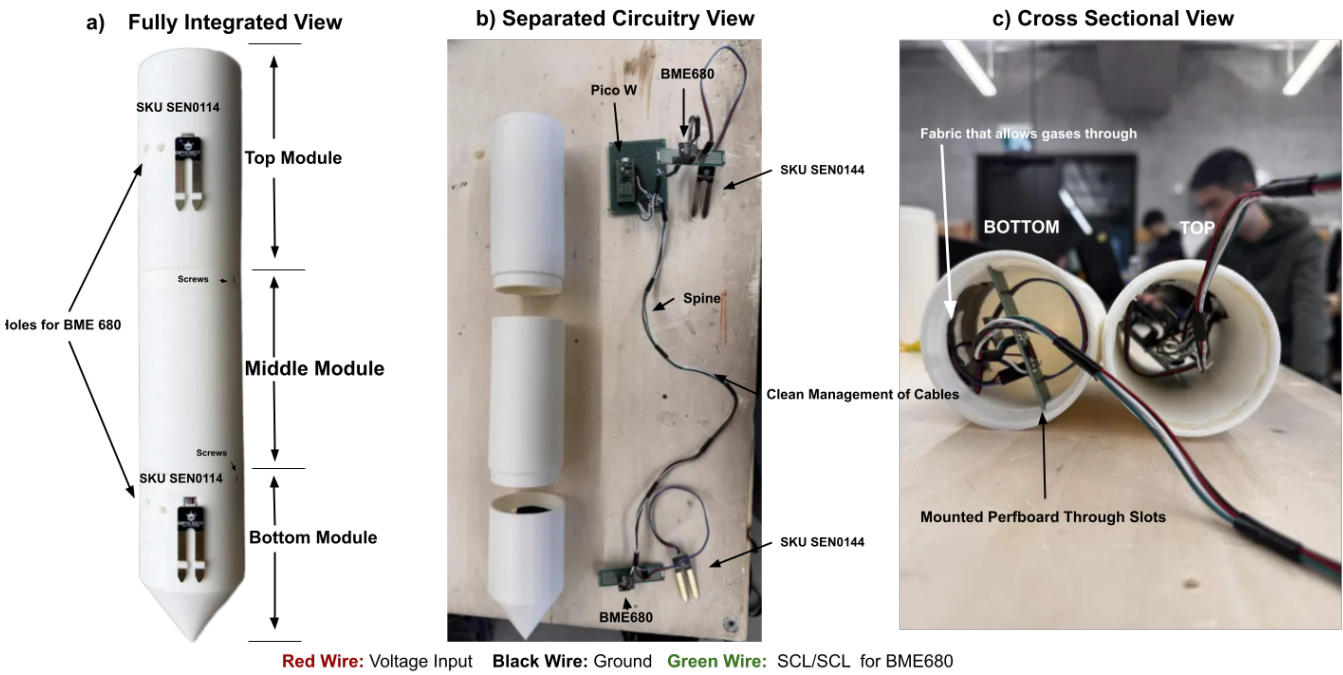


Figure 10: a) The fully integrated view shows the outer shell of the device, with components connected using screws. The SKU SEN0114 sensor is mounted. b) The internal circuitry illustrates how the sensors are connected. c) The cross-sectional view shows how the perfboards are mounted through internal slots. See figure 5 for Software System Architecture.

Our prototype successfully demonstrated that a ground sensor module can detect small changes in underground conditions, allowing communities, wildfire agencies, and municipalities to monitor subsurface environments more accurately.

1. Simple Modular Design:

The prototype has a simple, clean design consisting of two sensing modules and a central module that can accommodate a future battery pack. This configuration enables the measurement of gradient changes (S5), which is critical for monitoring zombie fires within a region. Future iterations could incorporate additional modules to expand sensing capabilities.

2. Easy Disassembly:

As shown in Figure 9b, the shells can be easily separated (S8). Combined with proper cable management and a simplified structural design, this allows for straightforward repair and reuse of the device, improving its environmental sustainability.

3. Future Steps:

To achieve a fully functional final design, the auger feature must be integrated, and long-distance communication capabilities will be required. As well, the insertion of a battery in the Middle Module is needed for long-term use.